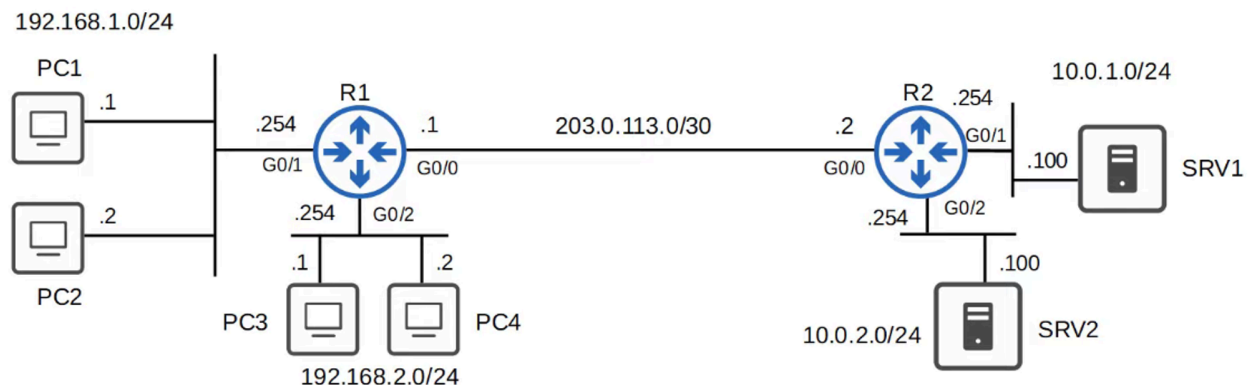


Day 34: Standard Access Control List (ACL)

WHAT ARE ACLs

- ACLs (Access Control Lists) have multiple uses
- In DAY 34 and DAY 35, we will focus on ACL's from a security perspective
- ACLs function as a "packet filter" - instructing the ROUTER to ALLOW or DENY specific traffic
- ACLs can filter traffic based on:
 - SOURCE / DESTINATION IP ADDRESSES
 - SOURCE / DESTINATION LAYER 4 PORTS
 - etc.

HOW ACLs WORK



💡 REQUIREMENTS:

- Hosts in 192.168.1.0/24 should have ACCESS to the 10.0.1.0/24 NETWORK
- Hosts in 192.168.2.0/24 should not have ACCESS to the 10.0.1.0/24 NETWORK

ACLs are configured GLOBALLY on the ROUTER (Global Config Mode)

- They are an ordered sequence of ACEs (Access Control Entries)

ACL 1:

- 1: if source IP = 192.168.1.0/24,
then permit
- 2: if source IP = 192.168.2.0/24,
then deny
- 3: if source IP = any, then permit

- Configuring an ACL in Global Config Mode will not make the ACL take effect
- The ACL must be applied to an interface
 - ACLs are applied either INBOUND or OUTBOUND
- ACLs are made up of one or more ACEs
- When a ROUTER checks a PACKET against the ACL, it processes the ACEs in order, from top to bottom
- If the PACKET matches one of the ACEs in the ACL, the ROUTER takes the action and stops processing the ACL. All entries below the matching entry will be ignored

ACL 2:

- 1: if source IP = 192.168.1.0/24,
then permit
- 2: if source IP = 192.168.0.0/16,
then deny

A maximum of one ACL can be applied to a single interface per direction.

Inbound: Maximum one ACL

Outbound: Maximum one ACL

IMPLICIT DENY

- What will happen if a PACKET doesn't match any of the entries in an ACL ?
- There is an INPLICIT DENY at the end of ALL ACL's
- The IMPLICIT DENY tells the ROUTER to DENY ALL TRAFFIC that doesn't match ANY of the configured entries in the ACL

ACL TYPES

- Standard ACLs: Match based on **Source IP address** only

→ Standard Numbered ACLs

→ Standard Named ACLs

- Extended ACLs: Match based on **Source/Destination IP, Source/Destination port**, etc.

→ Extended Numbered ACLs

→ Extended Named ACLs

STANDARD NUMBERED ACLs

- Match traffic based only on the SOURCE IP ADDRESS of the PACKET
- Numbered ACLs are identified with a number (ie: ACL 1, ACL 2, etc.)
- Different TYPES of ACLs have a different range of numbers that can be used
💡 STANDARD ACLs can use 1-99 and 1300-1999
- The basic command to configure a STANDARD NUMBERED ACL
 - `R1(config)# access-list *number* {deny | permit} *ip wildcard-mask*`

This is an example of denying a SPECIFIC host's traffic

REMEMBER : 0.0.0.0 wildcard is the same as 255.255.255.255 or a /32 host

- Example : `R1(config)# access-list 1 deny 1.1.1.1 0.0.0.0`
- Example : `R1(config)# access-list 1 deny 1.1.1.1` (identical to the above)
- Example : `R1(config)# access-list 1 deny host 1.1.1.1`

If you want to permit ANY traffic from ANY source

- Example : `R1(config)# access-list 1 permit any`
- Example : `R1(config)# access-list 1 permit 0.0.0.0 255.255.255.255`

If you want to make a description for a specific ACL

- Example : `R1(config)# access-list 1 remark ## BLOCK BOB FROM ACCOUNTING ##`

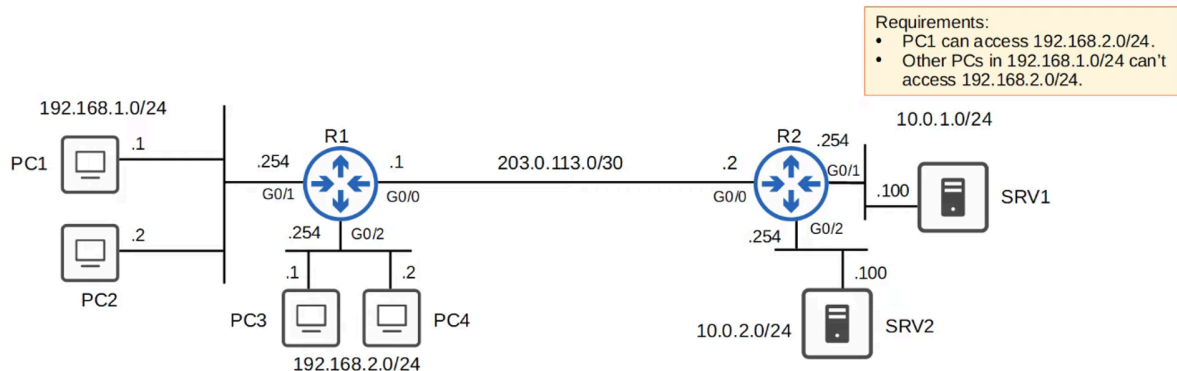
```
R1(config)#access-list 1 deny 1.1.1.1 0.0.0.0
R1(config)#access-list 1 permit 0.0.0.0 255.255.255.255
R1(config)#access-list 1 remark ## BLOCK BOB FROM ACCOUNTING ##
R1(config)#
R1(config)#do show access-lists
Standard IP access list 1
  10 deny 1.1.1.1
  20 permit any
R1(config)#
R1(config)#do show ip access-lists
Standard IP access list 1
  10 deny 1.1.1.1
  20 permit any
R1(config)#
R1(config)#do show running-config | include access-list
access-list 1 deny 1.1.1.1
access-list 1 permit any
access-list 1 remark ## BLOCK BOB FROM ACCOUNTING ##
R1(config)#
```

Order is important. Lower Numbers are processed FIRST

TO APPLY AN ACL TO AN INTERFACE

```
R1(config-if)# ip access-group *number* {in | out}
```

```
R1(config)#access-list 1 permit 192.168.1.1
R1(config)#access-list 1 deny 192.168.1.0 0.0.0.255
R1(config)#access-list 1 permit any
R1(config)#
R1(config)#interface g0/2
R1(config-if)#ip access-group 1 out
R1(config-if)#
```



WHY WAS THIS RULE PLACED ON G0/2 OUT ?

💡 STANDARD ACLs should be applied as CLOSE to the DESTINATION as possible!

STANDARD NAMED ACLs

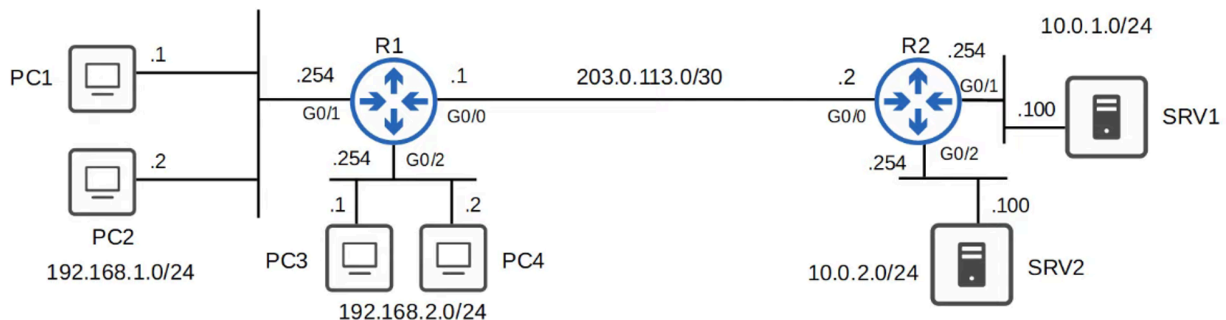
- Standard ACLs match traffic based only on the SOURCE IP ADDRESS of the PACKET
- NAMED ACLs are identified with a NAME (ie: 'BLOCK_BOB')
- STANDARD NAMED ACLs are configured by entering 'standard named ACL config mode' then configuring EACH entry within that config mode
 - `R1(config)# ip access-list standard *acl-name*`
 - `R1(config-std-nacl)# [*entry-number*] {deny | permit} *ip wildcard-mask*`

```
R1(config)#ip access-list standard BLOCK_BOB
R1(config-std-nacl)#5 deny 1.1.1.1
R1(config-std-nacl)#10 permit any
R1(config-std-nacl)#remark ## CONFIGURED NOV 21 2020 ##
R1(config-std-nacl)#interface g0/0
R1(config-if)#ip access-group BLOCK_BOB in
```

```
R1#show access-lists
Standard IP access list BLOCK_BOB
    5 deny    1.1.1.1
   10 permit any
R1#
R1#show running-config | section access-list
ip access-list standard BLOCK_BOB
    deny    1.1.1.1
    permit any
    remark ## CONFIGURED NOV 21 2020 ##
R1#
```

Requirements:

- PCs in 192.168.1.0/24 can't access 10.0.2.0/24.
- PC3 can't access 10.0.1.0/24.
- Other PCs in 192.168.2.0/24 can access 10.0.1.0/24.
- PC1 can access 10.0.1.0/24.
- Other PCs in 192.168.1.0/24 can't access 10.0.1.0/24.



Here are the configurations for the above:

```

R2(config)#ip access-list standard TO_10.0.2.0/24
R2(config-std-nacl)#deny 192.168.1.0 0.0.0.255
R2(config-std-nacl)#permit any
R2(config-std-nacl)#interface g0/2
R2(config-if)#ip access-group TO_10.0.2.0/24 out
R2(config-if)#
R2(config-if)#ip access-list standard TO_10.0.1.0/24
R2(config-std-nacl)#deny 192.168.2.1
R2(config-std-nacl)#permit 192.168.2.0 0.0.0.255
R2(config-std-nacl)#permit 192.168.1.1
R2(config-std-nacl)#deny 192.168.1.0 0.0.0.255
R2(config-std-nacl)#permit any
R2(config-std-nacl)#interface g0/1
R2(config-if)#ip access-group TO_10.0.1.0/24 out
R2(config-if)#

```

Note, however, how the order is when viewing the ACLs

```

R2#show ip access-lists
Standard IP access list TO_10.0.1.0/24
    30 permit 192.168.1.1
    10 deny    192.168.2.1
    20 permit 192.168.2.0, wildcard bits 0.0.0.255
    40 deny    192.168.1.0, wildcard bits 0.0.0.255
    50 permit any
Standard IP access list TO_10.0.2.0/24
    10 deny    192.168.1.0, wildcard bits 0.0.0.255
    20 permit any
R2#

```

WHY THE REORDERING?

- The router may re-order the /32 entries.
- This improves the efficiency of processing the ACL.
- It **does not** change the effect of the ACL.
- This applies to both standard named and standard numbered ACLs.
- Packet Tracer does not do this.

CISCOs PACKET TRACER does not reorder these, however.